INTEGRATION
Examination questions

1 Find a primitive function of $3x - 4$.

2 Find a primitive function of $16 - x^{-2}$.

3 Evaluate $\int_0^8 t^{\frac{1}{3}}\, dt$.

4 Find correct to one decimal place the value of
$$\int_0^2 \left(2x^5 + 1\right)\, dx.$$

5 Find $\int (3x-1)^5\, dx$.

6 Find $\int (x-1)^{\frac{1}{2}}\, dx$.

7 Find $\int_0^4 \left(2 - \sqrt{x}\right)\, dx$.

8 The curve $y = f(x)$ has gradient function
$$\frac{dy}{dx} = 3x^2 + 4x - 1.$$
The curve passes through the point $P(1, 0)$. Find its equation.

9 The graph of $y = f(x)$ passes through the point $(1, -1)$ and $f'(x) = 3x^2 + 2$. Find $f(x)$.

10 (i) Sketch the function $y = 2x^2$, $0 \le x \le 1$ and write down its domain and range.

(ii) If y in (i) is part of an odd function $f(x)$ defined for $-1 \le x \le 1$, sketch on a second diagram the function $f(x)$.

(iii) Write down the value of
$$\int_{-1}^{1} f(x)\, dx,$$
giving reasons for your answer.

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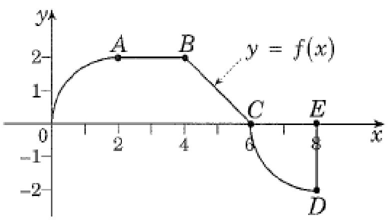


The graph of the function $f(x)$ consists of a quarter circle OA, a horizontal straight-line segment AB, a straight-line segment BC and a quarter circle CD, as shown above.

(i) Find the value of $\int_0^8 f(x)\, dx$.

(ii) Find the area under the curve $y = f(x)$ and the x-axis.

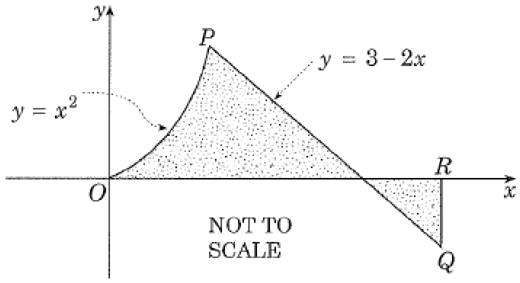
(iii) For what values of x lying between $0 < x < 8$ is the function not able to be differentiated?

12 Find the area between the curve
$$y = \frac{1}{(1-2x)^2},$$
the x-axis and the ordinates at $x = 0$ and $x = 2$.

13 (a) On a number plane, shade the region $\mathcal{R}$ bounded by the curve $y = 2x - x^2$ and the line $y = 3$, between $x = 0$ and $x = 2$.

(b) Find the area of $\mathcal{R}$.

14 The shaded region $OPQR$ is bounded by the lines $x = 0$, $x = 2$, the curve $y = x^2$, the line $y = 3 - 2x$ and the x-axis, as in the diagram.



(i) Find the coordinates of P.

(ii) What is the area of the shaded region $OPQR$?

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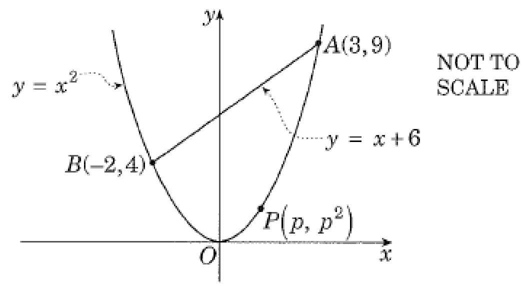


The points $A(3, 9)$ and $B(-2, 4)$ lie on the parabola $y = x^2$. The line $y = x + 6$ joins A and B. The point $P\left(p, p^2\right)$ is a variable point on the parabola below the line.

(i) Find the area of the parabolic segment APB, i.e. the area below the line and above the parabola.

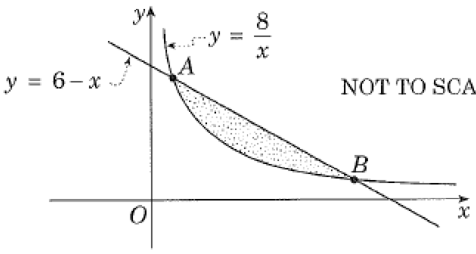
(ii) Show that the greatest possible area of the triangle APB is three-quarters of the area of the parabolic segment APB.

16 (i) Prove that the line $y = x - 4$ is a tangent to the parabola $y = x^2 - 5x + 5$.

(ii) Let P be the point where the line $y = x - 4$ touches the parabola $y = x^2 - 5x + 5$. Show that the normal to the parabola at P is $y = -x + 2$.

(iii) Find the area of the region enclosed between the parabola and the line $y = -x + 2$.

17



NOT TO SCALE

The line $y = 6 - x$ intersects the hyperbola $y = \frac{8}{x}$ at the points A and B, as shown above.

(i) Find the coordinates of the points A and B.

(ii) Find the area of the shaded region between $y = \frac{8}{x}$ and $y = 6 - x$.

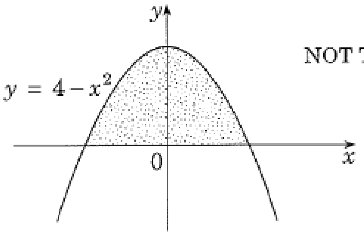
18 Find the volume of the solid of revolution formed by rotating the curve $y = x - \frac{1}{x}$ about the x-axis between $x = 1$ and $x = 4$.

19 (a) Find the points of intersection of the curve $y = 2 + \sqrt{4x}$ with the y-axis and the line $y = 4$.

(b) The area enclosed by the curve $y = 2 + \sqrt{4x}$, the line $y = 4$ and the y-axis is rotated about the y-axis. Find the volume of the solid of revolution formed.

20 The region bounded by the curve $y = 2\sqrt{x} - 1$, the lines $x = 3$, $x = 4$ and the x-axis, is rotated about the x-axis. Find the volume of the solid of revolution so formed.

21



NOT TO SCALE

The shaded region in the diagram is bounded by the curve $y = 4 - x^2$ and the x-axis is rotated about the x-axis.

Find the volume of the solid so formed.

22

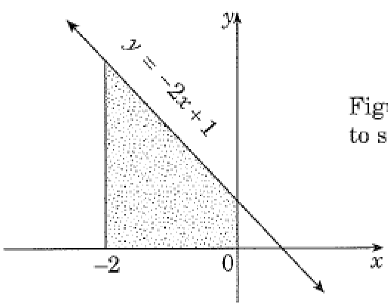
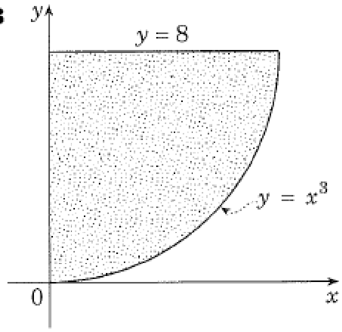


Figure not to scale

The shaded region in the diagram is bounded by the line $y = -2x + 1$, the x-axis, the y-axis, and the line $x = -2$.

Find the volume of revolution formed when this region is rotated about the x-axis.

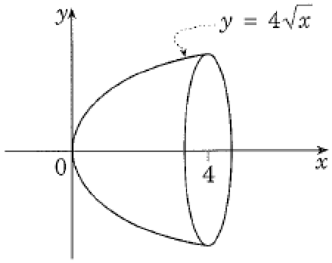
23



The shaded region shown at left is bounded by the curve $y = x^3$, the y-axis and the line $y = 8$.

Determine the volume of revolution formed when this region is rotated about the y-axis.

24



The region enclosed by the curve $y = 4\sqrt{x}$ and the x-axis between $x = 0$ and $x = 4$ is rotated about the x-axis.

Find the volume of the solid of revolution.

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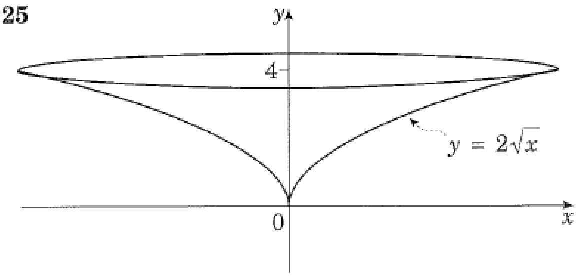
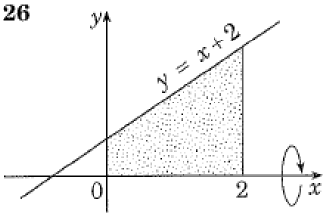


A curved funnel has a shape formed by rotating part of the parabola $y = 2\sqrt{x}$ about the y-axis as shown above. The funnel is 4 cm deep.

Find the volume of liquid which the funnel will hold if it is sealed at the bottom.

26



The shaded region formed between the x-axis and the line $y = x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis to form a solid.

Find the volume of the solid formed.

27 The function $f(x)$, defined for x in the domain $0 \le x \le 2$, is given by the rule

$$f(x) = (4-x)^{\frac{1}{3}}, \quad 0 \le x \le 2.$$

(a) Draw up a table of values of $f(x)$, correct to two decimal places, for

$$x = 0, \frac{1}{2}, 1, 1\frac{1}{2}, 2.$$

(b) Using the values of your table, draw a sketch of the graph of $y = f(x)$, for $0 \le x \le 2$. [Do not attempt to find any turning points.]

(c) Use the trapezoidal rule with four equal sub-intervals to estimate the area between the curve, the x-axis and the y-axis.

28 Consider the function $f(x) = \sqrt{16-x^2}$.

(i) Copy the following table of values into your writing booklet and calculate the missing values, correct to three decimal places.

x	0	1	2	3	4
$f(x)$	4·000		3·464		0·000

(ii) Use these five values of the function and the trapezoidal rule to find the approximate value of $\int_0^4 \sqrt{16-x^2}\,dx$.

(iii) Sketch the graph of $x^2 + y^2 = 16$ and shade the region whose area is represented by

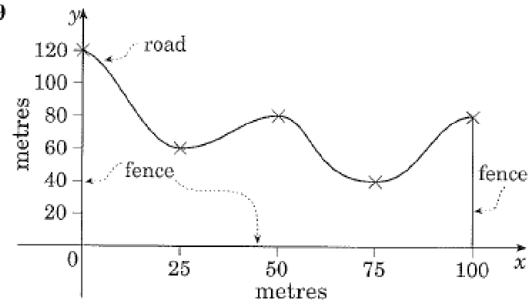
$$\int_0^4 \sqrt{16-x^2}\,dx.$$

(iv) By considering the answer of part (iii), explain why the exact area of

$$\int_0^4 \sqrt{16-x^2}\,dx \text{ is } 4\pi.$$

(v) By using your answers to parts (ii) and (iv), find an approximate value for π.

29



The diagram represents a scale drawing of a park bounded by a road and three fences.

The council surveyor wishes to calculate the area of the park and has measured perpendicular distances of the road from the bottom fence at intervals of 25 metres.

These distances can be read off the diagram.

(i) Take the bottom fence as the x-axis, the fences which are perpendicular to the bottom fence as the y-axis, and the line $x = 100$ and the road as $y = f(x)$.
Copy and complete the following table of values in your writing book.

x	0	25	50	75	100
$f(x)$					

(ii) Estimate the area of the park by using Simpson's rule with five function values.

30 The speed of a car was recorded at 2 minute intervals. The times, in minutes, and the corresponding speeds v, in kilometres per hour, appear in the table below.

time (min)	0	2	4	6	8
speed (km/h)	0	40	60	50	60

(i) Explain why the distance x, in km, travelled by the car in the 8 minutes is given by

$$x = \int_0^{\frac{2}{15}} v\,dt.$$

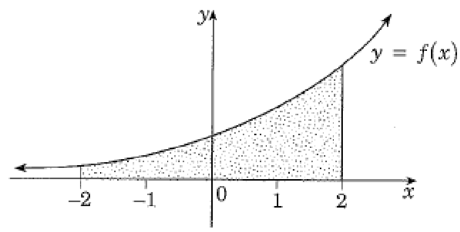
(ii) By using Simpson's rule with five function values, estimate the value of x.

31 Consider the function $y = 3^x$.

x	−2	−1	0	1	2
3^x					

(a) Copy and complete the above table in your writing book, leaving the values of $y = 3^x$ in fraction form.

(b) Use Simpson's rule with these five function values to find an estimate for the area shown shaded in the diagram below.



INTEGRATION
Worked solutions to examination questions

Remember the constant 'c'

1 $\int (3x-4)\,dx = \frac{3x^2}{2} - 4x + c$

2 $\int \left(16 - x^{-2}\right) dx = 16x - \frac{x^{-1}}{-1} + c$

$= 16x + x^{-1} + c$

$= 16x + \frac{1}{x} + c$

Remember: $\frac{1}{a/b} = \frac{b}{a} \Rightarrow \frac{1}{4/3} = \frac{3}{4}$

$8^{4/3} = \left(8^{1/3}\right)^4 = \left(\sqrt[3]{8}\right)^4 = 2^4$

3 $\int_0^8 t^{\frac{1}{3}}\,dt = \left[\frac{t^{4/3}}{4/3}\right]_0^8$

$= \frac{3}{4}\left(8^{4/3} - 0^{4/3}\right)$

$= \frac{3}{4}\left(2^4 - 0\right)$

$= \frac{3}{4} \times 16$

$= 12$

4 $\int_0^2 \left(2x^5 + 1\right) dx = \left[\frac{2x^6}{6} + x\right]_0^2$

$= \left(\frac{2^6}{3} + 2\right) - 0$

$= \frac{64}{3} + 2$

$= 23\frac{1}{3}$

$= 23{\cdot}3$ to 1 dec. place

Integration of a 'function of a function' type.

5 $\int (3x-1)^5\,dx = \frac{(3x-1)^6}{(6)(3)} + c$

$= \frac{(3x-1)^6}{18} + c$

6 $\int (x-1)^{1/2}\,dx = \frac{(x-1)^{3/2}}{(3/2)(1)} + c = \frac{2}{3}(x-1)^{3/2} + c$

Change $\sqrt{x}$ to $x^{1/2}$

7 $\int_0^4 \left(2 - \sqrt{x}\right) dx = \int_0^4 \left(2 - x^{1/2}\right) dx$

$= \left[2x - \frac{x^{3/2}}{3/2}\right]_0^4$

$= \left[2x - \frac{2}{3}\left(\sqrt{x}\right)^3\right]_0^4$

$= \left[2(4) - \frac{2}{3}\left(\sqrt{4}\right)^3\right] - 0$

$= 8 - \frac{2}{3}\left(2^3\right)$

$= 8 - \frac{16}{3}$

$= 8 - 5\frac{1}{3}$

$= 2\frac{2}{3}$

8 $\dfrac{dy}{dx} = 3x^2 + 4x - 1 \quad \therefore \quad y = \displaystyle\int \left(3x^2 + 4x - 1\right)\, dx$

$$= x^3 + 2x^2 - x + c$$

At the point (1, 0): $\quad 0 = 1^3 + 2\left(1^2\right) - 1 + c$

$$= 2 + c$$

$$\therefore \quad c = -2$$

$\therefore$ equation of the curve is $y = x^3 + 2x^2 - x - 2$

Ask yourself what process will reverse $\dfrac{dy}{dx}$ to obtain y.

9 $f'(x) = 3x^2 + 2 \quad \therefore \quad f(x) = \displaystyle\int \left(3x^2 + 2\right)\, dx$

$$= x^3 + 2x + c$$

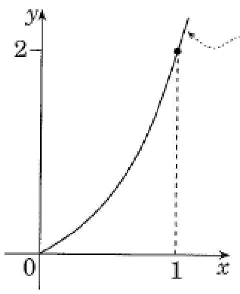
At the point (1, −1): $\quad -1 = 1^3 + 2(1) + c$

$$= 3 + c$$

$$\therefore \quad c = -4$$

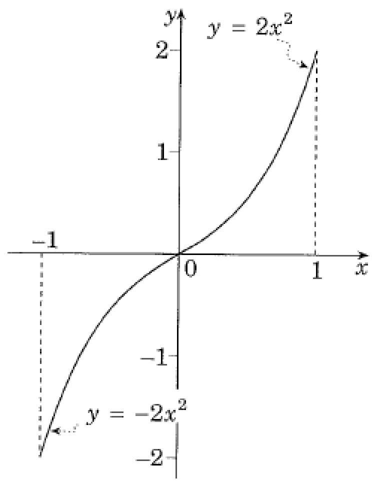
$\therefore$ equation of the curve is $f(x) = x^3 + 2x - 4$

10 (i)



Domain: $\{0 \le x \le 1\}$
Range: $\{0 \le y \le 2\}$

$y = 2x^2$
$= 2\left(1^2\right)$
$= 2$ when $x = 1$



(ii) A function is odd if $f(-x) = -f(x)$
For $-1 \le x \le 1$, the curve is shown by $y = -2x^2$

(ii) Think about the definition of an odd function.

(iii) $\displaystyle\int_{-1}^{1} f(x)\, dx = \int_{-1}^{0} f(x)\, dx + \int_{0}^{1} f(x)\, dx$

$$= -F(x) + F(x), \quad \text{where } F(x) = \int_0^1 f(x)\, dx$$

$$= 0$$

The value of $\displaystyle\int_{-1}^{0} f(x)\, dx$ is numerically the same, but opposite in sign to that of $\displaystyle\int_{0}^{1} f(x)\, dx$.

(iii) Make sure you understand the difference between finding the area under a curve and the value of an integral.

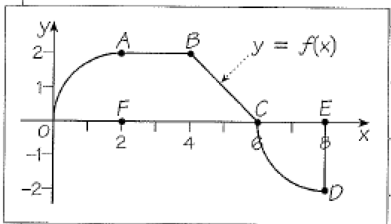
11 Let F be the point (2, 0)

(i) $\displaystyle\int_0^8 f(x)\, dx$ = area(quadrant OAF) + area(trapezium $FABC$) − area(quadrant CDE)

= area(trapezium $FABC$)

$= \frac{1}{2}(2)(2 + 4)$

$= 6$



(ii) Area under the curve and the x-axis

$= \text{area(quadrant } OAF) + \text{area(trapezium } FABC)$
$\quad + \text{area(quadrant } CDE)$

$= 6 + 2\left(\frac{1}{4} \cdot \pi \cdot 2^2\right)$

$= (6 + 2\pi) \text{ units}^2$

(iii) The curve is not differentiable at points B and C, i.e. at $x = 4$, $x = 6$

Note the difference for integrals and areas under a curve

Note: area(quadrant OAF) = area(quadrant CDE) (radius = 2)

12 Area $= \left| \int_0^2 \frac{dx}{(1-2x)^2} \right|$

$$= \left| \int_0^2 (1-2x)^{-2}\, dx \right|$$

$$= \left| \left[\frac{(1-2x)^{-1}}{(-1)(-2)} \right]_0^2 \right|$$

$$= \left| \frac{1}{2}\left[\frac{1}{1-2x} \right]_0^2 \right|$$

$$= \left| \frac{1}{2}\left(\frac{1}{1-4} - \frac{1}{1-0} \right) \right|$$

$$= \left| \frac{1}{2}\left(\frac{1}{-3} - 1 \right) \right|$$

$$= \left| \frac{1}{2}\left(-\frac{4}{3} \right) \right|$$

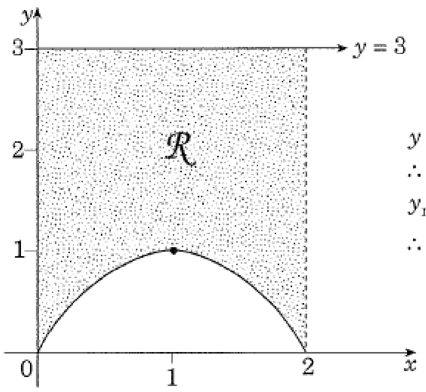
$$= \left| -\frac{2}{3} \right|$$

$$= \frac{2}{3} \text{ units}^2$$

12 Use absolute value for the area, as we may not know in advance if the area is above or below the x-axis.

Change $\frac{1}{(1-2x)^2}$ to negative index form.

13 (a)



$y = 2x - x^2 = x(2-x)$

$\therefore$ x-intercepts at $x = 0, 2$

$y_{\max}$ at $x = 1$

$\therefore$ $y_{\max} = 2(1) - 1^2 = 1$

(b) Area(shaded) = area(below line $y = 3$) − area(under parabola)

$\therefore$ area($\mathcal{R}$) $= \int_0^2 \left(y_{\text{upper}} - y_{\text{lower}} \right) dx$

$= \int_0^2 \left[3 - \left(2x - x^2 \right) \right] dx$ where $y_{\text{upper}} = 3$, $y_{\text{lower}} = 2 - x^2$

$= \int_0^2 \left(3 - 2x + x^2 \right) dx$

continued . . .

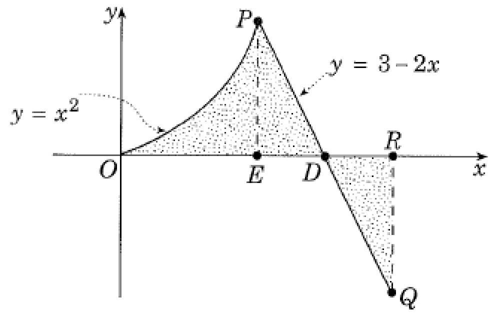
$$= \left[3x - x^2 + \frac{x^3}{3}\right]_0^2$$

$$= \left[3(2) - 2^2 + \frac{2^3}{3}\right] - 0$$

$$= 6 - 4 + \frac{8}{3}$$

$$= 4\tfrac{2}{3} \text{ units}^2$$

14



14 Intersection of two curves
⇒ solve simultaneously

(i) Solving $y = x^2$ and $y = 3 - 2x$ simultaneously to find the point P,

we have $\qquad x^2 = 3 - 2x$

$\therefore \qquad x^2 + 2x - 3 = 0$

$\qquad (x-1)(x+3) = 0$

$\therefore \quad x = 1, -3$

From the diagram, it is obvious that $x = 1$ gives the point P.

At $x = 1$, $\qquad y = x^2 = 1^2 = 1$

$\therefore$ P has coordinates $(1, 1)$

(ii) Let E, D be points on the x-axis, as shown.

E directly below the point P has the abscissa $x = 1$

$\therefore$ E has coordinates $(1, 0)$

At $y = 0$, $\qquad 3 - 2x = 0$

$\therefore \qquad 2x = 3$

$\qquad x = 1\frac{1}{2}$

$\therefore$ D has coordinates $\left(1\frac{1}{2}, 0\right)$

R is the point $(2, 0)$

At $x = 2$, the line $y = 3 - 2x$

$\qquad = 3 - 2(2)$

$\qquad = -1$

$\therefore$ Q has coordinates $(2, -1)$

Area (shaded) = area (OEP) + area (ΔEPD) + area (ΔDQR)

$$= \int_0^1 x^2\,dx + \tfrac{1}{2}.ED.EP + \tfrac{1}{2}.DR.RQ$$

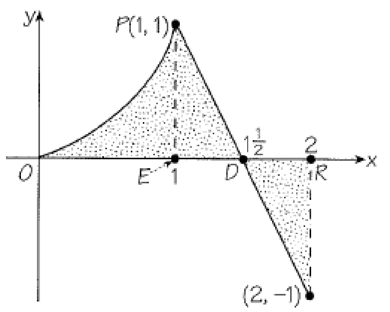
$$= \left[\frac{x^3}{3}\right]_0^1 + \tfrac{1}{2}.\tfrac{1}{2}.1 + \tfrac{1}{2}.\tfrac{1}{2}.|-1|$$

$$= \left(\tfrac{1}{3} - 0\right) + \tfrac{1}{4} + \tfrac{1}{4}$$

$$= \tfrac{1}{3} + \tfrac{1}{2}$$

$$= \tfrac{5}{6} \text{ units}^2$$

Split the shape into separate areas.



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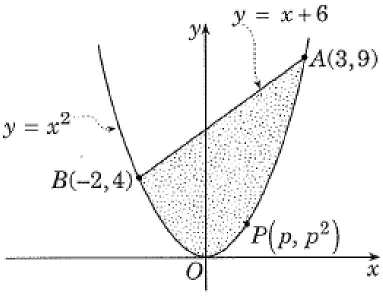


(i) Area (shaded) $= \int \left(y_{\text{upper}} - y_{\text{lower}}\right) dx$

$$= \int_{-2}^{3} \left[(x+6) - x^2\right] dx$$

$$= \int_{-2}^{3} \left(x + 6 - x^2\right) dx$$

$$= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3}\right]_{-2}^{3}$$

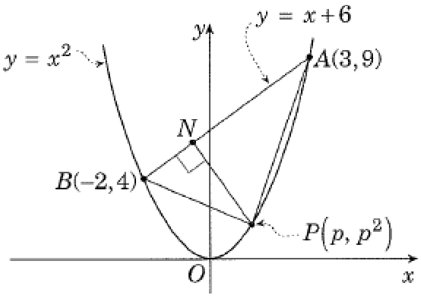
$$= \left[\frac{3^2}{2} + 6(3) - \frac{3^3}{3}\right] - \left[\frac{(-2)^2}{2} + 6(-2) - \frac{(-2)^3}{3}\right]$$

$$= \left(\frac{9}{2} + 18 - 9\right) - \left(2 - 12 + \frac{8}{3}\right)$$

$$= 13\tfrac{1}{2} - \left(-7\tfrac{1}{3}\right)$$

$$= 20\tfrac{5}{6} \text{ units}^2$$

(ii)



Let N be the foot of the perpendicular from P to AB.
The line AB has equation $y = x + 6$, i.e. $x - y + 6 = 0$

Distance $PN = \left|\dfrac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$

$\therefore \quad PN = \left|\dfrac{1(p) - 1\left(p^2\right) + 6}{\sqrt{1^2 + (-1)^2}}\right|$ with $x_1 = p$, $y_1 = p^2$

$$= \left|\frac{p - p^2 + 6}{\sqrt{2}}\right|$$

Distance $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\therefore \quad AB = \sqrt{[3 - (-2)]^2 + (9 - 4)^2}$

$$= \sqrt{5^2 + 5^2}$$
$$= \sqrt{50}$$
$$= 5\sqrt{2}$$

(ii) On your diagram form up the ΔAPB. Mark in the perpendicular height.
Consider the distance from a point (P) to a line (AB).
Find the area of ΔAPB (in terms of p).

$\text{Area}(\Delta APB) = \frac{1}{2}.AB.PN$

i.e. $\quad A = \frac{1}{2}.5\sqrt{2}\left(\frac{p-p^2+6}{\sqrt{2}}\right)$ for $-2 \le p \le 3$

$$= \frac{5}{2}\left(p-p^2+6\right)$$

$$\frac{dA}{dp} = \frac{5}{2}(1-2p) = 0 \text{ for stationary values}$$

i.e. $\quad 1-2p = 0$

$\therefore \quad p = \frac{1}{2}$

Using the 2nd derivative test for max/min

$$\frac{d^2A}{dp^2} = \frac{5}{2}(-2)$$

$= -5 < 0 \quad \therefore$ A is a maximum at $p = \frac{1}{2}$

$$\begin{aligned}\text{Area}(\Delta APB) &= \frac{5}{2}\left(p-p^2+6\right)\\ &= \frac{5}{2}\left(\frac{1}{2}-\frac{1}{2^2}+6\right)\\ &= \frac{5}{2}\left(6\frac{1}{4}\right)\\ &= \frac{125}{8}\\ &= 15\frac{5}{8}\text{ units}^2\end{aligned}$$

$$\begin{aligned}\frac{\text{area}(\Delta APB)}{\text{area}(\text{segment } APB)} &= \frac{15\frac{5}{8}}{20\frac{5}{6}}\\ &= \frac{125}{8}\times\frac{6}{125}\\ &= \frac{3}{4}\end{aligned}$$

$\therefore$ maximum area $\Delta APB = \frac{3}{4}\times$ area(segment APB)

16 (i) To find the point of contact of the line $y = x-4$ and the parabola $y = x^2-5x+5$, solve simultaneously.

$\therefore \quad x-4 = x^2-5x+5$

i.e. $\quad x^2-6x+9 = 0$

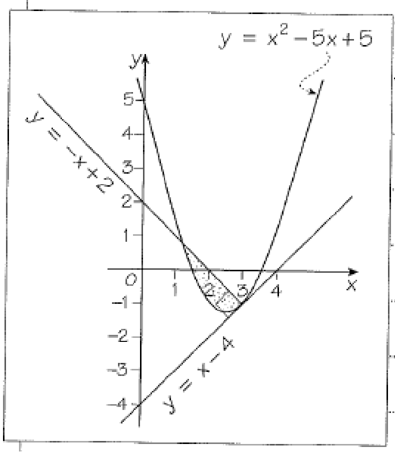
$(x-3)^2 = 0$

$\therefore \quad x = 3$

At $x = 3$, $\quad \begin{aligned} y &= x-4\\ &= 3-4\\ &= -1\end{aligned}$

$\therefore$ the point of contact is $(3, -1)$.

Since there is only one point of contact, the line must be a tangent to the curve.



(ii) P is the point of contact, as in (i)

$\therefore$ P has coordinates $(3, -1)$

Gradient of $y = x-4$ is 1.

Gradient of normal to $y = x-4$ is -1 (using $m_1m_2 = -1$)

Equation of normal $\frac{y-y_1}{x-x_1} = m$: $\quad \frac{y-(-1)}{x-3} = -1$

$\therefore \quad y+1 = -x+3$

i.e. $\quad y = -x+2$

(iii) The line $y = -x+2$ and the parabola $y = x^2-5x+5$ meet at the points found by solving simultanously.

$\therefore \quad x^2-5x+5 = -x+2$

i.e. $\quad x^2-4x+3 = 0$

$(x-3)(x-1) = 0$

$\therefore \; x = 1$ and 3

Area (between line and parabola)

$$= \int_1^3 \left(y_{\text{upper}} - y_{\text{lower}}\right)\, dx$$

$$= \int_1^3 \left[(-x+2)-\left(x^2-5x+5\right)\right] dx$$

$$= \int_1^3 \left(-x+2-x^2+5x-5\right) dx$$

$$= \int_1^3 \left(4x-x^2-3\right) dx$$

$$= \left[2x^2-\frac{x^3}{3}-3x\right]_1^3$$

$$= \left[2\left(3^2\right)-\frac{3^3}{3}-3(3)\right]-\left[2\left(1^2\right)-\frac{1^3}{3}-3(1)\right]$$

$$= (18-9-9)-\left(2-\tfrac{1}{3}-3\right)$$

$$= 0-\left(-1\tfrac{1}{3}\right)$$

$$= 1\tfrac{1}{3} \text{ units}^2$$

17 (i) The coordinates of A and B are found by solving $y = \dfrac{8}{x}$ and $y = 6-x$ simultaneously.

$$\therefore \quad 6-x = \frac{8}{x}$$

$$6x-x^2 = 8$$

$$x^2-6x+8 = 0$$

$$(x-2)(x-4) = 0$$

$\therefore \; x = 2$ and 4

At $x = 2$, $\quad y = \dfrac{8}{x} = \dfrac{8}{2} = 4$

At $x = 4$, $\quad y = \dfrac{8}{x} = \dfrac{8}{4} = 2$

$\therefore$ the coordinates are $A(2, 4)$ and $B(4, 2)$.

(ii) Area (shaded) $= \int \left(y_{\text{upper}} - y_{\text{lower}}\right)\, dx$

$$= \int_2^4 \left[(6-x)-\frac{8}{x}\right] dx$$

$$= \int_2^4 \left(6-x-\frac{8}{x}\right) dx$$

$$= \left[6x-\frac{x^2}{2}-8\ln x\right]_2^4$$

$$= \left[6(4)-\frac{4^2}{2}-8\ln 4\right]-\left[6(2)-\frac{2^2}{2}-\ln 2\right]$$

$$= (24-8-8\ln 4)-(12-2-\ln 2)$$

To be attempted after logarithms/exponentials have been taught.

continued . . .

$= 16 - 8\ln 4 - 10 + \ln 2$
$= 6 - 16\ln 2 + \ln 2$
$= (6 - 15\ln 2)$ units^2

$8\ln 4 = 8\ln 2^2$
$= 8 \times 2\ln 2$
$= 16\ln 2$

18 $V = \pi\int y^2\,dx = \pi\int_1^4\left(x - \frac{1}{x}\right)^2 dx$

$$= \pi\int_1^4\left(x^2 - 2 + \frac{1}{x^2}\right) dx$$

$$= \pi\int_1^4\left(x^2 + x^{-2} - 2\right) dx$$

$$= \pi\left[\frac{x^3}{3} + \frac{x^{-1}}{-1} + 2x\right]_1^4$$

$$= \pi\left[\frac{x^3}{3} - \frac{1}{x} + 2x\right]_1^4$$

$$= \pi\left\{\left[\frac{4^3}{3} - \frac{1}{4} + 2(4)\right] - \left(\frac{1}{3} - 1 + 2\right)\right\}$$

$$= \pi\left(\frac{64}{3} - \frac{1}{4} + 8 - \frac{1}{3} + 1 - 2\right)$$

$$= \pi\left(\frac{63}{3} + 7 - \frac{1}{4}\right)$$

$$= \pi\left(21 + 7 - \frac{1}{4}\right)$$

$$= \pi\left(28 - \frac{1}{4}\right)$$

$$= \pi\left(27\tfrac{3}{4}\right)$$

$$= \frac{111\pi}{4} \text{ units}^3$$

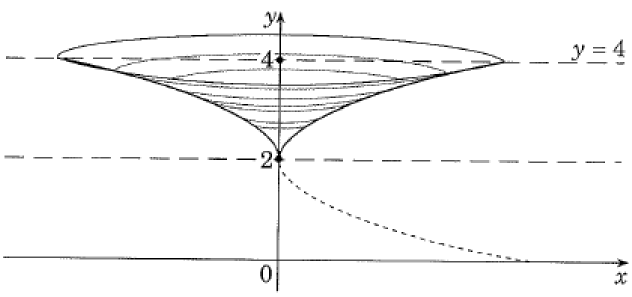
19 (a) On the y-axis, $x = 0$:

$y = 2 + \sqrt{4x}$
$\therefore\ y = 2 + \sqrt{0}$
i.e. $y = 2$

Point of intersection of the parabola and the line $y = 4$:

$y = 2 + \sqrt{4x}$
$\therefore\ 4 = 2 + \sqrt{4x}$
$2 = \sqrt{4x}$
$2^2 = 4x$
$\therefore\ x = 1$

(b)



$y = 2 + \sqrt{4x}$
$y - 2 = \sqrt{4x}$
$(y-2)^2 = 4x \Rightarrow$ half parabola, vertex $(0, 2)$

From $(y-2)^2 = 4x, \quad x = \dfrac{(y-2)^2}{4}$

$$V = \pi \int x^2 \, dy$$
$$= \pi \int_2^4 \left[\frac{(y-2)^2}{4}\right]^2 dy$$
$$= \frac{\pi}{16} \int_2^4 (y-2)^4 \, dy$$
$$= \frac{\pi}{16} \left[\frac{(y-2)^5}{5}\right]_2^4$$
$$= \frac{\pi}{80}\left[(4-2)^5 - (2-2)^5\right]$$
$$= \frac{\pi}{80} \times 2^5$$
$$= \frac{32\pi}{80}$$
$$= \frac{2\pi}{5} \text{ units}^3$$

About the y-axis

$$\Rightarrow V = \pi \int x^2 \, \boxed{dy}$$

20 $V = \pi \int y^2 \, dx = \pi \int_3^4 \left(2\sqrt{x} - 1\right)^2 dx$

$$= \pi \int_3^4 \left(4x - 4\sqrt{x} + 1\right) dx$$
$$= \pi \int_3^4 \left(4x - 4x^{1/2} + 1\right) dx$$
$$= \pi \left[2x^2 - \frac{4x^{3/2}}{(3/2)} + x\right]_3^4$$
$$= \pi \left[2x^2 - \frac{8}{3} x\sqrt{x} + x\right]_3^4$$
$$= \pi \left\{\left[2\left(4^2\right) - \frac{8}{3}.4\sqrt{4} + 4\right] - \left(2.3^2 - \frac{4}{3}.3\sqrt{3} + 3\right)\right\}$$
$$= \pi \left[\left(32 - \frac{32}{3}.2 + 4\right) - \left(18 - 4\sqrt{3} + 3\right)\right]$$
$$= \pi\left(36 - 21\tfrac{1}{3} - 21 + 4\sqrt{3}\right)$$
$$= \pi\left(4\sqrt{3} - 6\tfrac{1}{3}\right)$$
$$= \pi\left(4\sqrt{3} - \frac{19}{3}\right) \text{ units}^3$$

$\boxed{x^{3/2} = x \cdot x^{1/2} = x\sqrt{x}}$

21 The parabola $y = 4 - x^2$ intersects the x-axis at $y = 0$

$\therefore \quad 0 = 4 - x^2$

$x^2 = 4$

$x = \pm\sqrt{4} = \pm 2$

i.e. at $x = -2$ and $x = 2$

$$V = \pi \int y^2 \, dx \text{ about the } x\text{-axis}$$
$$= \pi \int_{-2}^2 \left(4 - x^2\right)^2 dx$$

continued . . .

$$= \pi\int_{-2}^{2}\left(16-8x^2+x^4\right)\,dx$$

$$= \pi\left[16x-\frac{8x^3}{3}+\frac{x^5}{5}\right]_{-2}^{2}$$

$$= 2\pi\left[16x-\frac{8x^3}{3}+\frac{x^5}{5}\right]_{0}^{2}$$

$$= 2\pi\left[16(2)-\frac{8(2)^3}{3}+\frac{(2)^5}{5}\right]-0$$

$$= 2\pi\left(32-\frac{64}{3}+\frac{32}{5}\right)$$

$$= 2\pi\left(32-21\tfrac{1}{3}+6\tfrac{2}{5}\right)$$

$$= 2\pi\left(17\tfrac{1}{15}\right)$$

$$= 2\pi\left(\frac{256}{15}\right)$$

$$= \frac{512\pi}{15}\ \text{units}^3$$

If $f(x)$ is even,

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

22 $V = \pi\int y^2\,dx$ about the x-axis

$$= \pi\int_{-2}^{0}(-2x+1)^2\,dx$$

$$= \pi\int_{-2}^{0}\left(4x^2-4x+1\right)dx$$

$$= \pi\left[\frac{4x^3}{3}-2x^2+x\right]_{-2}^{0}$$

$$= \pi\left\{0-\left[\frac{4(-2)^3}{3}-2(-2)^2+(-2)\right]\right\}$$

$$= \pi\left[\frac{32}{3}+8+2\right]$$

$$= \pi\left(20\tfrac{2}{3}\right)$$

$$= \frac{62\pi}{3}\ \text{units}^3$$

23 $V = \pi\int x^2\,dy$ about the y-axis

$$= \pi\int_{0}^{8} y^{2/3}\,dy \qquad \text{since } x^3 = y \Rightarrow x = \sqrt[3]{y} = y^{1/3}$$

$$x^2 = y^{2/3}$$

$$= \pi\left[\frac{y^{5/3}}{5/3}\right]_0^8$$

$$= \pi\cdot\frac{3}{5}\left[y^{5/3}\right]_0^8$$

$$= \frac{3\pi}{5}\left[\left(\sqrt[3]{y}\right)^5\right]_0^8$$

$$= \frac{3\pi}{5}\left[\left(\sqrt[3]{8}\right)^5\right]$$

$$= \frac{3\pi}{5}\cdot\left(2^5\right) = \frac{3\pi}{5}\cdot 32 = \frac{96\pi}{5}\ \text{units}^3$$

24 $V = \pi\int y^2\ dx = \pi\int_0^4 \left(4\sqrt{x}\right)^2 dx$

$$= \pi\int_0^4 16x\ dx$$

$$= 16\pi\left[\frac{x^2}{2}\right]_0^4$$

$$= 8\pi\left[4^2 - 0\right]$$

$$= 8\pi(16)$$

$$= 128\pi \text{ units}^3$$

25 $y = 2\sqrt{x} \Rightarrow \frac{y}{2} = \sqrt{x} \qquad \therefore\ x = \frac{y^2}{4}$

$V = \pi\int x^2\ dy$ about the y-axis

$$= \pi\int_0^4 \left(\frac{y^2}{4}\right)^2 dy$$

$$= \pi\int_0^4 \frac{y^4}{16}\ dy$$

$$= \frac{\pi}{16}\left[\frac{y^5}{5}\right]_0^4$$

$$= \frac{\pi}{80}\cdot 1024$$

$$= \frac{64\pi}{5} \text{ units}^3$$

26 $V = \pi\int y^2\ dx$ about the x-axis

$$= \pi\int_0^2 (x+2)^2\ dx$$

$$= \pi\left[\frac{(x+2)^3}{3}\right]_0^2$$

$$= \frac{\pi}{3}\left(4^3 - 2^3\right)$$

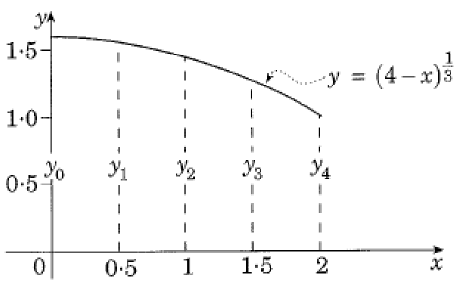
$$= \frac{\pi}{3}(64 - 8)$$

$$= \frac{56\pi}{3} \text{ units}^3$$

27 (a) $f(x) = (4-x)^{\frac{1}{3}} = \sqrt[3]{(4-x)}$ for $0 \le x \le 2$

x	0	0·5	1·0	1·5	2·0
$f(x)$	1·59	1·52	1·44	1·36	1·26

(b)



(c) Using the trapezoidal rule, we have

$$A \approx \frac{h}{2}(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4)$$
$$\approx \frac{0{\cdot}5}{2}[1{\cdot}59 + 2(1{\cdot}52) + 2(1{\cdot}44) + 2(1{\cdot}36) + 1{\cdot}26]$$
$$= 0{\cdot}25(11{\cdot}49)$$
$$= 2{\cdot}87 \text{ units}^2$$

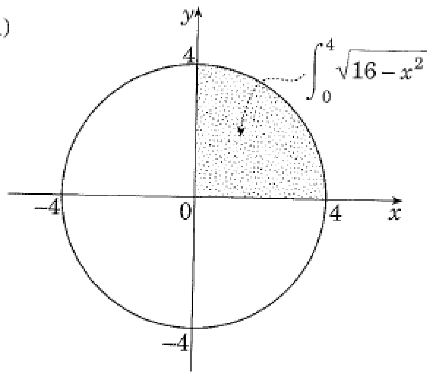
28 (i) $f(x) = \sqrt{16 - x^2}$

x	0	1	2	3	4
$f(x)$	4·000	3·873	3·464	2·646	0·000

(ii) Using the trapezoidal rule, we have

$$A \approx \frac{h}{2}(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4), \qquad h = 1$$
$$= \frac{h}{2}[y_0 + 2(y_1 + y_2 + y_3) + y_4]$$
$$= \frac{1}{2}[4 + 2(3{\cdot}873 + 3{\cdot}464 + 2{\cdot}646) + 0]$$
$$= \frac{1}{2}(23{\cdot}966)$$
$$\approx 11{\cdot}983 \text{ units}^2$$

(iii)



$x^2 + y^2 = 16$
$\Rightarrow$ circle, centre $(0, 0)$
radius $= 4$

$y = \sqrt{16 - x^2}$
$\Rightarrow$ quarter circle in the first quadrant

(iv) $\int_0^4 \sqrt{16 - x^2}\ dx$ represents the area of a quarter circle, radius $= 4$

Area circle $= \pi r^2 = \pi(4^2) = 16\pi \text{ units}^2$

$$\therefore \int_0^4 \sqrt{16 - x^2}\ dx = \frac{1}{4}\cdot(16\pi) = 4\pi \text{ units}^2$$

(v) Since the exact value is approximately equal to the result using the trapezoidal rule, we have

$$11{\cdot}983 \approx 4\pi$$
$$\therefore \quad \pi \approx \frac{11{\cdot}983}{4}$$
i.e. $\quad \pi \approx 2{\cdot}996$

29 (i) Reading y-values from the diagram, we have

x	0	25	50	75	100
$f(x)$	120	60	80	40	100
	y_0	y_1	y_2	y_3	y_4

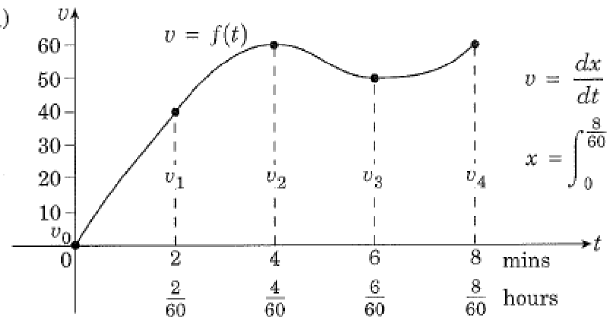
$h = 25$

(ii) By Simpson's rule, $A \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + y_4)$

$$\approx \frac{25}{3}[120 + 4(60) + 2(80) + 4(40) + 100]$$
$$= \frac{25}{3}(780)$$
$$= 6500 \text{ m}^2$$

> **30** To be attempted after completing 'Applications of calculus to the physical world'.

30 (i)



$v = \frac{dx}{dt}$ by definition

$$x = \int_0^{\frac{8}{60}} v\, dt = \int_0^{\frac{2}{15}} v\, dt$$

> $8 \text{ min} = \frac{8}{60} = \frac{2}{15}$ hours

$x = \int_0^{\frac{2}{15}} v\, dt$ is equivalent to the area under the curve $v = f(t)$, from $t = 0$ to $t = \frac{2}{15}$ hour.

(ii) Using Simpson's rule,

$$A \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + y_4)$$

i.e. $x \approx \frac{h}{3}(v_0 + 4v_1 + 2v_2 + 4v_3 + v_4)$

$$= \frac{(1/30)}{3}[0 + 4(40) + 2(60) + 4(50) + 60]$$
$$= \frac{1}{90}(540)$$
$$= 6 \text{ km}$$

> $h = 2 \text{ min} = \frac{2}{60} = \frac{1}{30}$ hour

31 $y = 3^x$

(a)

x	-2	-1	0	1	2
y	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9
	y_0	y_1	y_2	y_3	y_4

$h = 1$

(b) $A \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + y_4)$

$$\approx \frac{1}{3}\left[\frac{1}{9} + 4\left(\frac{1}{3}\right) + 2(1) + 4(3) + 9\right]$$
$$= \frac{1}{3}\left(\frac{1}{9} + \frac{4}{3} + 23\right)$$
$$= \frac{1}{3}\left(24\tfrac{4}{9}\right)$$
$$= 8\tfrac{4}{27} \text{ units}^2$$